

Conditional Mean Reversion in U.S. Equities: Machine Learning, Regime Dependence, and Transaction Costs

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Abstract

We study whether nonlinear machine-learning models can improve next-month cross-sectional U.S. equity return forecasts, and whether those forecasts survive a disciplined long-short implementation after transaction costs. The empirical design uses only course-provided Monthly CRSP and Compustat-style data, with a final audited expanding-window walk-forward protocol that evaluates annual out-of-sample test folds from 2017 through 2024. The main specification combines return features with an audited flow-only Compustat block and maps monthly predictions into equal-weight top-minus-bottom decile portfolios. The strongest result is the flow-Compustat MLP, which delivers net Sharpe 2.03 at 10 bps, annualized net return 21.4%, a rank-IC HAC t -statistic of 6.36, and breakeven costs near 90 bps. The strategy strongly outperforms same-protocol reversal and z -score reversal benchmarks. Momentum remains the strongest classical comparator; the ML strategies are economically stronger, but mean-difference tests versus momentum are not strongly significant. Performance is higher in bear and high-volatility states, yet direct regime-feature ablations do not improve the strong flow-Compustat models. The central limitation is that the provided Monthly CRSP extract omits `prc`, `shrout`, `shrcd`, and `exchcd`, so the report supports a credible flat-cost academic result rather than a full tradability proof.

Introduction

Short-term reversal is one of the oldest cross-sectional return patterns in equity markets, but it is not obviously stable enough to trade in raw form. Naive reversal portfolios are high-turnover, sensitive to market conditions, and often strongest in parts of the universe where implementation is hardest. The more relevant question for a machine-learning project is therefore not whether reversal exists mechanically, but whether conditional expected returns can be estimated well enough to rank stocks more profitably than simple fixed rules.

This project treats machine learning as a cross-sectional forecasting and ranking problem. Each month, the model observes lagged stock-level return signals, optionally augments them with audited flow-only firm characteristics, and predicts next-month

returns. Those forecasts are then converted into market-neutral long-short portfolios and evaluated under a shared backtest engine with flat transaction costs. The resulting claim is economic rather than purely statistical: ML is useful only if it improves out-of-sample ranking and retains performance after realistic turnover penalties.

The report makes five contributions. First, it uses a final audited expanding-window protocol designed to prevent leakage and align all model comparisons. Second, it compares nonlinear models against both linear benchmarks and same-protocol classical reversal/momentum strategies. Third, it studies cost sensitivity through turnover and breakeven transaction-cost analysis. Fourth, it treats regime dependence both as an evaluation dimension and as a direct feature ablation question. Fifth, it states clearly what cannot be claimed under the assignment’s no-external-data rule.

Data and Prediction Problem

The empirical target comes from Monthly CRSP. For stock i at month t , the main label is next-month return,

$$R_{i,t+1} = \text{target_ret_fwd_1m}.$$

The report focuses on this raw return target. A supplementary robustness exercise replaces it with market-adjusted next-month return, `target_excess_ret_fwd_1m`, but that does not replace the main result.

Predictors come from two admissible sources within the course data. The first is a return-feature block constructed from lagged CRSP information. The second is an audited flow-only Compustat block merged point-in-time with conservative availability rules. No external datasets are used. The sample spans 1984-01-31 through 2024-11-30, while the out-of-sample test window covers 2017-2024. The walk-forward design keeps the training start fixed at 1985-01-31 and rolls forward by calendar test year.

Table 1 summarizes the sample. Two additional facts from the feature audit matter for interpretation. First, the Compustat integration reports zero point-in-time violations. Second, roughly 66.8% of panel

Statistic	Value
Sample period	1984-01-31 to 2024-11-30
Stock-months	3,646,731
Unique stocks	33,884
Average stocks per month	7,427
Baseline return features	15
Main target	target_ret_fwd_1m
Target mean	0.85%
Target volatility	18.39%

Table 1: Sample overview used in the final pipeline.

Source: table_data_summary.csv in report/tables/.

rows have at least one flow-only Compustat field available, which is enough to matter economically without redefining the whole sample around a small subset of firms.

The largest scope constraint is a missing-data one rather than a modeling one. The provided Monthly CRSP extract lacks `prc`, `shrout`, `shrcd`, and `exchcd`. Under the assignment’s no-external-data rule, that means the project cannot implement true price, share-code, exchange, microcap, or liquidity screens. The correct interpretation is therefore flat-cost robustness on the provided universe, not a full executable tradability study.

Features, Models, and Validation Design

The baseline feature family, `return_only`, is built entirely from lagged CRSP returns. It includes recent returns, intermediate-horizon momentum, trailing volatility, drawdown, market beta, idiosyncratic volatility, and cross-sectional rank transforms. These variables are designed to let the model learn where reversal is strong, weak, or overturned by other return dynamics.

The main feature family, `flow_compustat`, augments those return signals with audited flow-only firm characteristics such as sales, cost of goods sold, capital expenditure, research and development, SG&A, margins, growth rates, and intensity ratios. These are the final Compustat inputs because the reviewed extract supports them with defensible point-in-time timing. Balance sheet style ratios are not part of the active pipeline because they are not reliably available in the provided schema.

The model comparison has three layers. The classical baselines are reversal, z-score reversal, momentum, and reversal plus momentum. The historical linear benchmarks are Ridge and Elastic Net. The nonlinear models are XGBoost and a compact MLP trained with regularization and early stopping. The report centers on the final hierarchy in which the main comparison set is return-only XGBoost, return-only

MLP, flow-Compustat XGBoost, and flow-Compustat MLP.

Validation is strict. For each annual test fold $Y \in \{2017, \dots, 2024\}$, training uses data from 1985-01-31 through $Y - 6$, validation uses the next five years $Y - 5$ through $Y - 1$, and the held-out calendar year Y is the test period. Hyperparameters are selected on validation data only. Targets are shifted forward by one month, predictors are available at time t , and only out-of-sample predictions enter the backtest. The detailed ML-only comparison is recorded in `table_walkforward_model_summary.csv`; it shows that the linear models remain materially weaker than the nonlinear specifications.

Preprocessing is fold-specific as well. Continuous predictors are winsorized and rank-normalized using training information only, missing values are imputed without using future observations, and missingness indicators are retained when they add information. That discipline matters because the Compustat block is informative partly through selective availability, and any full-sample transformation would contaminate the walk-forward evaluation.

Portfolio Construction and Transaction Costs

The forecasting problem is evaluated as a ranking problem. Each month, model scores are converted into cross-sectional ranks, and the strategy goes long the top 10% of names and short the bottom 10%. Long and short sleeves are equal-weighted, so gross leverage is 2 and net market exposure is approximately zero by construction. Alternative 5% and 20% cutoffs are part of the robustness suite, but the 10% specification is the headline reporting standard.

Transaction costs are modeled as a flat basis-point charge on turnover:

$$R_{p,t+1}^{\text{net}} = R_{p,t+1}^{\text{gross}} - c \times \text{turnover}_t,$$

where c is the round-trip cost expressed in decimal form. The main table uses 10 bps, and robustness considers a wider grid from zero to 100 bps. This cost model is intentionally simple. It is useful for comparing signals on a common footing, but it does not substitute for a full liquidity-aware execution model, especially given the missing tradability fields described above.

The shared portfolio engine also sharpens interpretation. Every strategy is judged under the same decile cutoff, equal-weighting rule, rebalance frequency, and cost convention, so performance differences can be attributed primarily to signal quality rather than to inconsistent construction choices. That is especially important when comparing ML models with classical

Strategy	Role	Net SR	Gross SR	Net ann.	Turnover	Breakeven	Rank-IC t
Flow-Compustat $\times$ MLP	Main model	2.028	2.281	21.4%	2.21	90.5	6.36
Flow-Compustat $\times$ XGBoost	ML comparison	1.770	1.987	21.3%	2.18	91.4	3.54
Return-only $\times$ XGBoost	ML comparison	1.313	1.531	18.0%	2.50	70.1	2.09
Return-only $\times$ MLP	ML comparison	1.192	1.512	12.3%	2.76	47.2	4.04
Momentum	Strongest classical benchmark	0.636	0.683	15.0%	0.92	145.6	3.34
Reversal plus momentum	Combined benchmark	0.491	0.662	8.2%	2.40	38.6	2.74
Reversal	Classical benchmark	0.062	0.224	1.4%	3.13	13.9	-0.12
Z-score reversal	Classical benchmark	0.062	0.224	1.4%	3.13	13.9	-0.12

Table 2: Final model hierarchy under the final audited expanding-window walk-forward protocol. *Source:* `table_final_model_hierarchy.csv` in `report/tables/`.

Feature set	Model	Gross SR	Net SR	Net ann.	Turnover	Breakeven	Rank-IC t
Flow-Compustat	MLP	2.281	2.028	21.4%	2.21	90.5	6.36
Flow-Compustat	XGBoost	1.987	1.770	21.3%	2.18	91.4	3.54
Return-only	XGBoost	1.531	1.313	18.0%	2.50	70.1	2.09
Return-only	MLP	1.512	1.192	12.3%	2.76	47.2	4.04
Flow-Compustat	Elastic Net	0.438	0.237	3.2%	2.25	21.8	0.99
Flow-Compustat	Ridge	0.234	-0.005	-0.1%	2.08	9.8	1.36
Return-only	Elastic Net	0.088	-0.109	-1.8%	2.66	4.5	0.69
Return-only	Ridge	-0.162	-0.284	-6.4%	2.29	-13.3	-0.72

Table 3: Detailed ML-only walk-forward comparison. *Source:* `table_walkforward_model_summary.csv` in `report/tables/`.

baselines, because raw forecast scale matters less than cross-sectional ranking power.

Results

Table 2 is the central economic ranking of strategies. It combines the detailed walk-forward model table with the same-protocol classical baselines and focuses on the strategies that define the report’s main story.

Four conclusions stand out. First, nonlinear models dominate the linear benchmarks and the raw reversal rules. The strongest specification is the flow-Compustat MLP, which combines net Sharpe 2.03 with gross Sharpe 2.28 and annualized net return 21.4% at 10 bps. Second, audited firm-level flow features materially improve the nonlinear models. Relative to the return-only MLP, the flow-Compustat MLP gains roughly 84 basis points of monthly mean spread, lifts net Sharpe from 1.19 to 2.03, and raises breakeven costs from 47 to 90 bps. The same direction holds for XGBoost, though less strongly.

Table 3 makes the linear-versus-nonlinear gap explicit. Even after the audited flow-only Compustat extension, the linear models are economically weak, with flow-Compustat Elastic Net at net Sharpe 0.24 and flow-Compustat Ridge effectively zero after costs. The return-only linear models are outright negative net at 10 bps. That matters because it rules out a weaker interpretation in which any modest predictive fit would survive the portfolio step; the gain is concentrated in the nonlinear architectures.

Third, the classical comparison is nuanced. The ML strategies strongly beat reversal and z-score reversal both economically and statistically, which is important because those are the natural mean-reversion

benchmarks. Momentum is more competitive. Its net Sharpe of 0.64 is still well below the two leading ML strategies, but momentum turns over much less and retains a very high breakeven cost. The correct claim is therefore that ML clearly dominates reversal, while momentum remains a meaningful classical benchmark rather than a straw man.

The incremental-value tests in `table_model_incremental_tests.csv` clarify where the Compustat lift matters most. The flow-Compustat MLP beats the return-only MLP by about 0.75% per month, or 9.05% annualized, with Newey-West $t = 2.53$. The analogous XGBoost lift is positive but much smaller, about 3.24% annualized with $t = 0.90$. The difference between the two flow-Compustat nonlinear winners is economically negligible. This pattern fits the interpretation that audited firm-flow variables are most valuable when the model can use them to learn nonlinear interactions rather than merely adding them linearly.

Fourth, the stronger ML models remain robust to nontrivial flat costs. `table_cost_breakeven.csv` shows that the flow-Compustat MLP and flow-Compustat XGBoost still have positive net Sharpe at 75 bps, about 0.39 and 0.36 respectively, and only turn slightly negative near 100 bps. By contrast, the weaker specifications lose economic viability far earlier.

Figure 1 makes the same point visually. The cumulative return panel shows a clear separation between the leading flow-Compustat models and the rest of the field. The cost panel shows that the main economic claim is not driven by an unrealistically low 10 bps assumption. Flat-cost robustness is not the same as

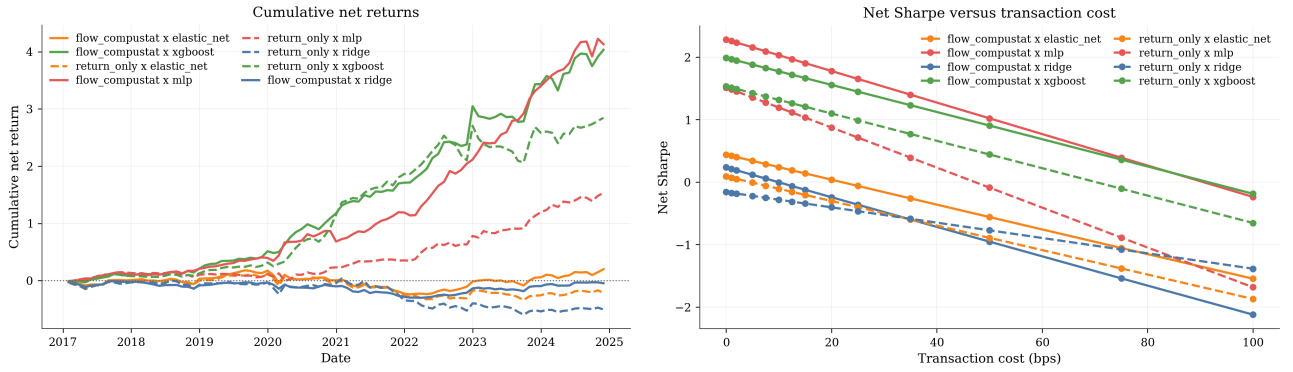


Figure 1: Cumulative net performance and transaction-cost sensitivity. The left panel shows that the leading nonlinear specifications accumulate economic gains through the final test window. The right panel shows that the main flow-Compustat models remain positive under materially higher flat costs than the weaker alternatives. *Source:* cumulative-return and cost-sensitivity figures in `report/figures/`; cost values align with `table_cost_breakeven.csv`.

Panel A: same-protocol classical baselines	Net SR	Net ann.	Turnover	Breakeven	Rank-IC t
Momentum	0.636	15.0%	0.92	145.6	3.34
Reversal plus momentum	0.491	8.2%	2.40	38.6	2.74
Reversal	0.062	1.4%	3.13	13.9	-0.12
Z-score reversal	0.062	1.4%	3.13	13.9	-0.12

Panel B: selected mean-difference tests versus classical baselines	Months	Monthly diff.	Annualized diff.	NW t	Interpretation
Flow-Compustat MLP – reversal	95	1.66%	19.93%	3.34	Strong outperformance
Flow-Compustat XGBoost – reversal	95	1.65%	19.82%	4.28	Strong outperformance
Flow-Compustat MLP – z-score reversal	95	1.66%	19.93%	3.34	Strong outperformance
Flow-Compustat XGBoost – z-score reversal	95	1.65%	19.82%	4.28	Strong outperformance
Flow-Compustat MLP – momentum	95	0.53%	6.35%	0.74	Economic, not decisive
Flow-Compustat XGBoost – momentum	95	0.52%	6.25%	0.56	Economic, not decisive

Table 4: Classical benchmark levels and same-protocol model-difference tests. *Source:* same-protocol classical baseline and model-difference CSVs in `report/tables/`.

implementation proof, but it is enough to rule out fragile paper alpha.

Robustness and Ablations

Table 4 sharpens the interpretation of the classical benchmark comparison. Reversal and z-score reversal are weak under the same walk-forward protocol. Momentum is stronger and should be treated as the real classical hurdle. The Newey-West mean-difference tests support exactly that reading: both leading ML strategies strongly outperform reversal, while the economic edge over momentum is not statistically decisive in this sample. Same-protocol classical baselines close the main benchmark gap, and momentum is a stronger classical benchmark than reversal.

The other robustness dimensions are also informative. Table 5 summarizes the key implementation and exposure diagnostics. Rebalancing results from `table_rebalance_robustness.csv` show that quarterly trading cuts average turnover from roughly 2.2 to 0.86 for the leading flow-Compustat models, but monthly rebalancing still produces the stronger net Sharpe. Alpha and beta estimates from

`table_alpha_beta.csv` show that the main MLP and XGBoost strategies earn positive net monthly alpha of about 1.98% and 1.89%, respectively, with low or negative market beta. These results are consistent with a cross-sectional stock-selection effect rather than a disguised market bet.

Prediction sorting also looks economically credible. Figure 2 shows clear separation between the tails of the forecast distribution. The flow-Compustat MLP delivers a gross top-minus-bottom spread of about 2.00% per month, with decile 1 averaging -1.02% and decile 10 averaging 1.05%. The flow-Compustat XGBoost model is similar, at roughly 1.99% per month. Monotonicity is strongest at the tails rather than perfectly smooth in every intermediate decile, which is typical for noisy cross-sectional return data.

Regime dependence is present in performance even when it does not help as an explicit feature input. According to `table_regime_robustness.csv`, the flow-Compustat MLP reaches net Sharpe 3.41 in bear months and 2.56 in high-volatility months, while still remaining positive in bull and low-volatility states. The corresponding XGBoost model reaches 1.95 in bear months and 2.23 in high-volatility months. This

Panel A: incremental-value tests	Monthly diff.	Annualized diff.	NW t
Flow-Compustat MLP – Return-only MLP	0.75%	9.05%	2.53
Flow-Compustat XGBoost – Return-only XGBoost	0.27%	3.24%	0.90
Flow-Compustat MLP – Flow-Compustat XGBoost	0.01%	0.10%	0.03
Panel B: implementation and exposure diagnostics			
Strategy	Monthly / quarterly net SR	Monthly / quarterly turnover	Net alpha / beta
Flow-Compustat MLP	2.03 / 1.85	2.21 / 0.86	1.98% / -0.17
Flow-Compustat XGBoost	1.77 / 1.37	2.18 / 0.86	1.89% / -0.10

Table 5: Selected incremental-value, implementation, and exposure diagnostics. *Source:* `table_model_incremental_tests.csv`, `table_rebalance_robustness.csv`, and `table_alpha_beta.csv` in `report/tables/`.

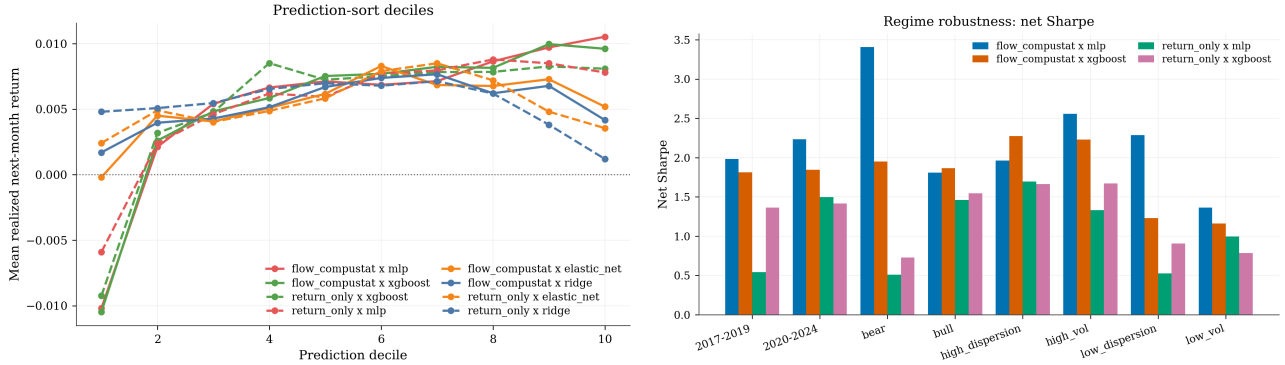


Figure 2: Prediction-decile evidence and regime-conditional net Sharpe. The left panel shows that the best models sort the tails well. The right panel shows that performance varies materially across market states, with the strongest outcomes in bear and high-volatility regimes. *Source:* `figure_prediction_deciles.png` and `figure_regime_net_sharpe.png` in `report/figures/`; supporting values align with the prediction-decile and regime-robustness CSVs in `report/tables/`.

supports the interpretation that mean reversion is conditional and state-dependent.

Subperiod robustness points in the same direction. Table 6 summarizes the regime and subperiod net Sharpe evidence for the two leading models. The flow-Compustat MLP posts net Sharpe 1.98 in 2017-2019 and 2.23 in 2020-2024, while the corresponding XGBoost specification delivers 1.81 and 1.84. High-dispersion months are especially favorable for XGBoost, at net Sharpe 2.27, whereas low-dispersion and low-volatility states remain positive but weaker for both leading models. The signal is therefore not a one-episode artifact, but it is stronger when cross-sectional opportunities widen.

However, the direct regime-feature ablation is not supportive for the best models. Table 7 shows that adding a small lagged regime block to the flow-Compustat models lowers net Sharpe from 2.028 to 1.520 for the MLP and from 1.770 to 1.439 for XGBoost. The return-only MLP improves modestly when regime features are added, but that gain is not enough to challenge the main model. The right interpretation is that regime dependence matters for evaluation, but simple regime features are not a robust incremental input to the strongest specification. We therefore treat regime dependence mainly as an evaluation dimension.

The remaining supplementary results are favorable. Re-estimating the main flow-Compustat MLP on the excess-return target yields net Sharpe 2.97 and breakeven costs near 124 bps. A three-seed MLP robustness check gives mean net Sharpe 1.97 with standard deviation 0.23, minimum 1.72, and maximum 2.16. These results reinforce the baseline finding, but they should remain supplementary because the headline result is the final audited raw-return walk-forward result.

These checks help bound the interpretation of the main result. The excess-target exercise shows that the signal is not simply a proxy for market direction, and the multi-seed exercise shows that the main MLP is not a single favorable initialization. At the same time, neither robustness layer changes the hierarchy of evidence: the paper's primary claim continues to rest on the raw-return final audited walk-forward package because that is the protocol used for the final comparison set. These checks do not replace the main raw-target final audited result.

Limitations

The main limitation is implementation realism. The report uses a shared flat-bps transaction-cost model and the provided Monthly CRSP universe, but it cannot apply true price, share-code, exchange, microcap,

Regime or subperiod	Flow-Compustat MLP net SR	Flow-Compustat XGBoost net SR
2017-2019	1.981	1.813
2020-2024	2.232	1.843
Bear market months	3.408	1.952
Bull market months	1.808	1.863
High-volatility months	2.559	2.229
Low-volatility months	1.361	1.161
High-dispersion months	1.962	2.273
Low-dispersion months	2.286	1.229

Table 6: Selected regime-conditional and subperiod net Sharpe results for the two leading strategies. *Source:* `table_regime_robustness.csv` in `report/tables/`.

Panel A: supplementary target and seed checks	Net SR	Breakeven	Rank-IC t	Mean seed SR	Seed std.
Flow-Compustat MLP on excess-return target	2.970	124.0	4.51	–	–
Flow-Compustat MLP multi-seed stability	–	–	–	1.968	0.227

Panel B: direct regime-feature ablation at 10 bps

Specification	Net SR	Reference SR	Delta SR	Target	Interpretation
Flow-Compustat-regime $\times$ MLP	1.520	2.028	-0.508	Raw return	Weaker than main model
Flow-Compustat-regime $\times$ XGBoost	1.439	1.770	-0.332	Raw return	Weaker than flow-Compustat XGBoost
Return-regime $\times$ MLP	1.398	1.192	0.207	Raw return	Modest help to return-only MLP
Return-regime $\times$ XGBoost	1.267	1.313	-0.046	Raw return	Roughly flat to weaker

Table 7: Supplementary robustness checks and direct regime-feature ablations. These results support, but do not replace, the final audited raw-return walk-forward result. *Source:* `table_supplementary_robustness.csv` in `report/tables/`.

or liquidity screens because the required fields are absent from the provided extract. That means the paper demonstrates robust academic signal quality under a controlled backtest rather than a fully executable production strategy.

The second limitation is that monthly data compresses trading frictions into a single turnover penalty. There is no explicit market-impact model, no order-book information, and no richer execution schedule. Likewise, delisting return handling is limited to what is already present in the provided source data. These constraints are acceptable within the assignment but should narrow the strength of any external investment claim.

A related limitation is that the regime block is deliberately simple. The report uses lagged market-return, volatility, drawdown, and dispersion proxies rather than a full macro state model. That is appropriate for a disciplined course project, but it means the regime results should be read as evidence of conditional heterogeneity rather than as a complete map from macro states to expected reversal profits.

The final limitation is conceptual. Even with strictly out-of-sample evaluation, ML finance results remain exposed to model-selection risk and multiple-testing risk. The protocol protects the 2017-2024 test folds from direct leakage, but it does not make the broader research process immune to data-mining concerns. For that reason, the report’s strongest claim is comparative and conditional: within the provided-data environment, nonlinear models with audited flow-Compustat inputs rank stocks better than the natural

reversal benchmarks.

Conclusion

The final evidence supports a clear but measured conclusion. Conditional nonlinear models improve cross-sectional mean-reversion forecasting relative to simple reversal rules, and the improvement survives a common 10 bps cost assumption. The strongest specification is the flow-Compustat MLP, with net Sharpe 2.03, annualized net return 21.4%, breakeven costs near 90 bps, and strong rank-IC evidence. Flow-Compustat XGBoost is the closest competitor and confirms that the result is not tied to a single nonlinear architecture.

The conclusion becomes more nuanced once the correct classical benchmark is used. ML strongly dominates reversal and z-score reversal, but momentum remains economically relevant and is not cleanly rejected by the same-protocol mean-difference tests. Regime dependence also matters, but mainly as a source of performance heterogeneity rather than as an explicit feature block that improves the strongest model. Overall, the project provides credible out-of-sample evidence that conditional mean reversion is strongest when nonlinear methods are combined with firm-level flow characteristics, while also making clear that the result remains an academic ML-in-finance finding rather than a complete tradability proof.